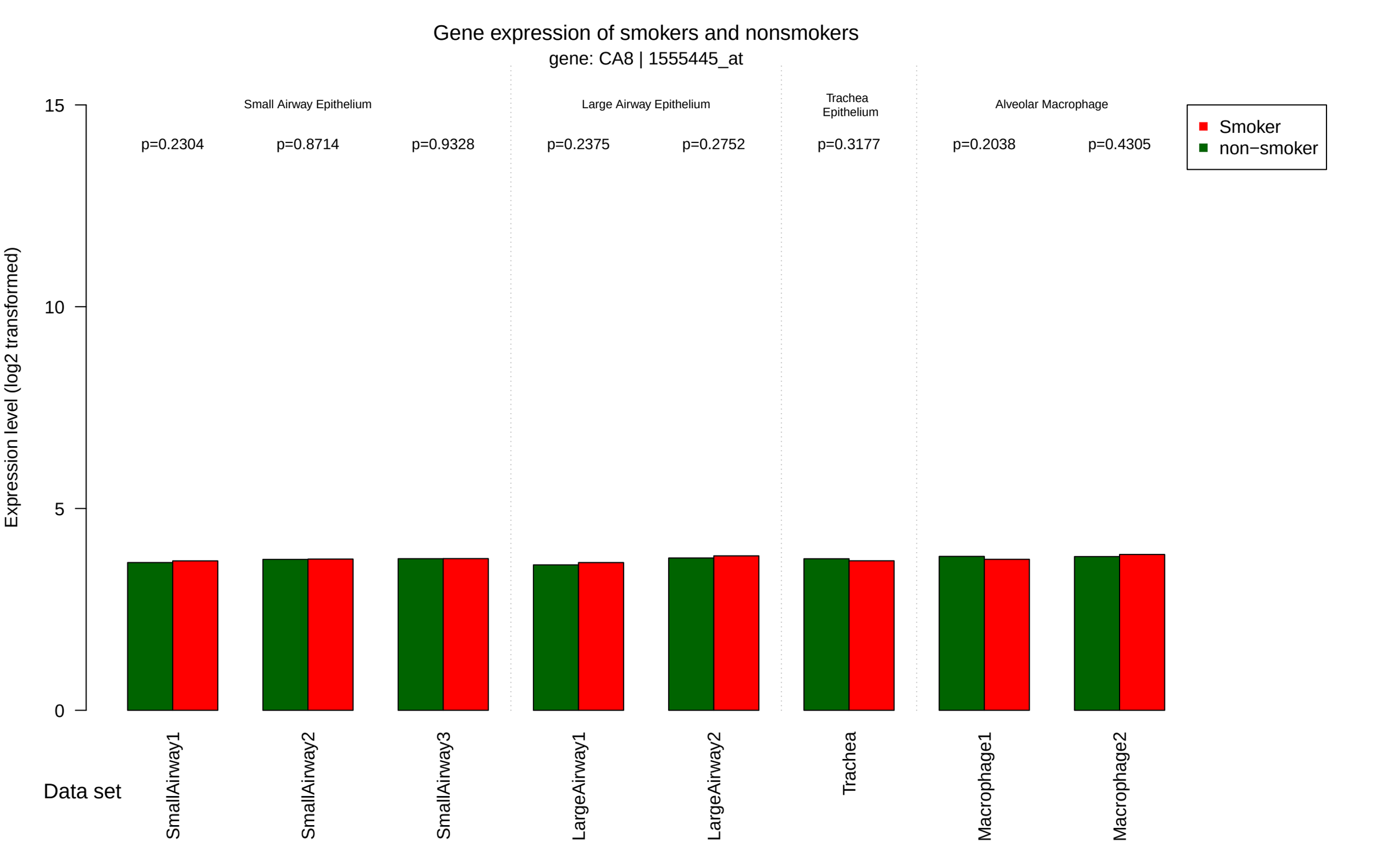
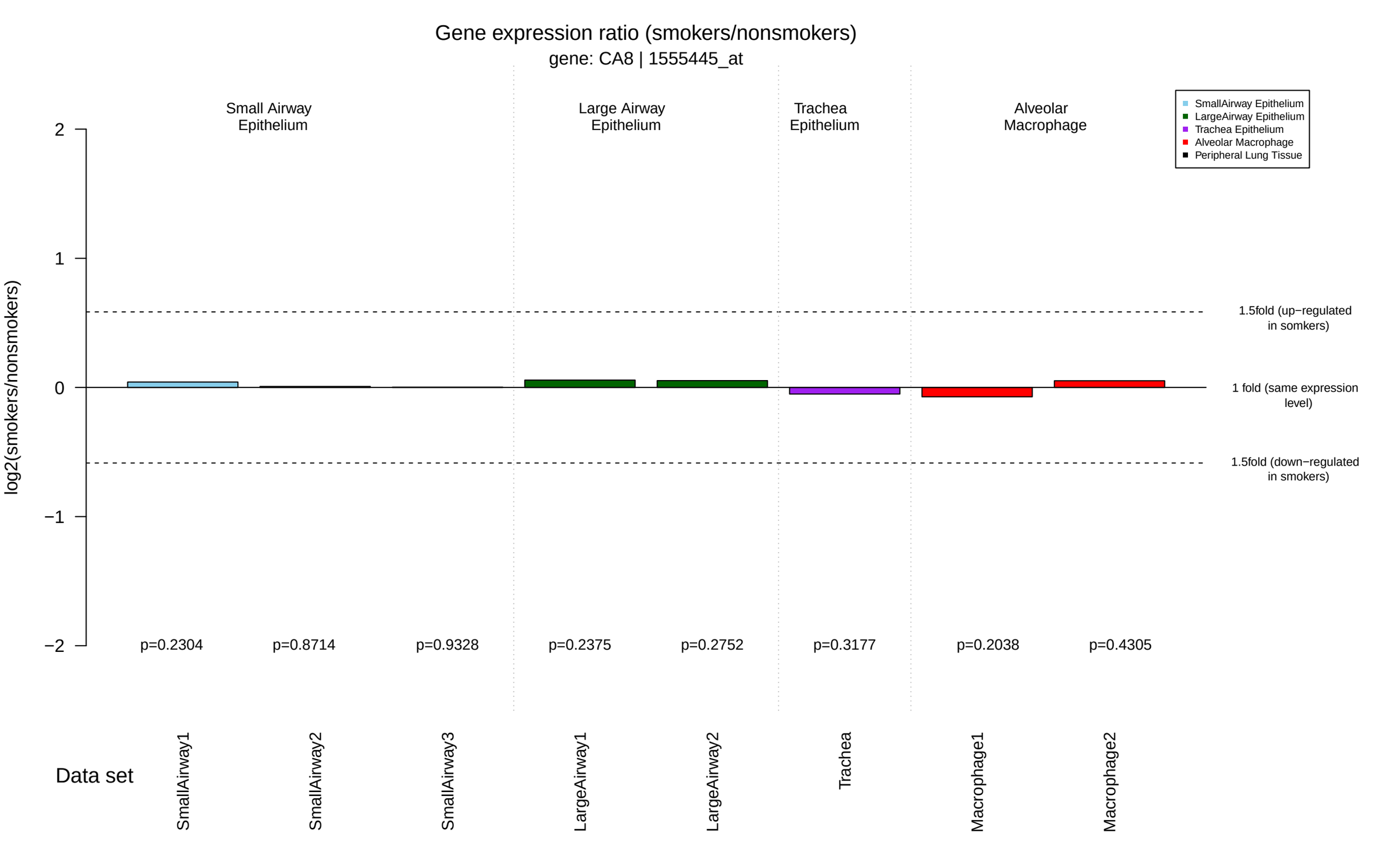
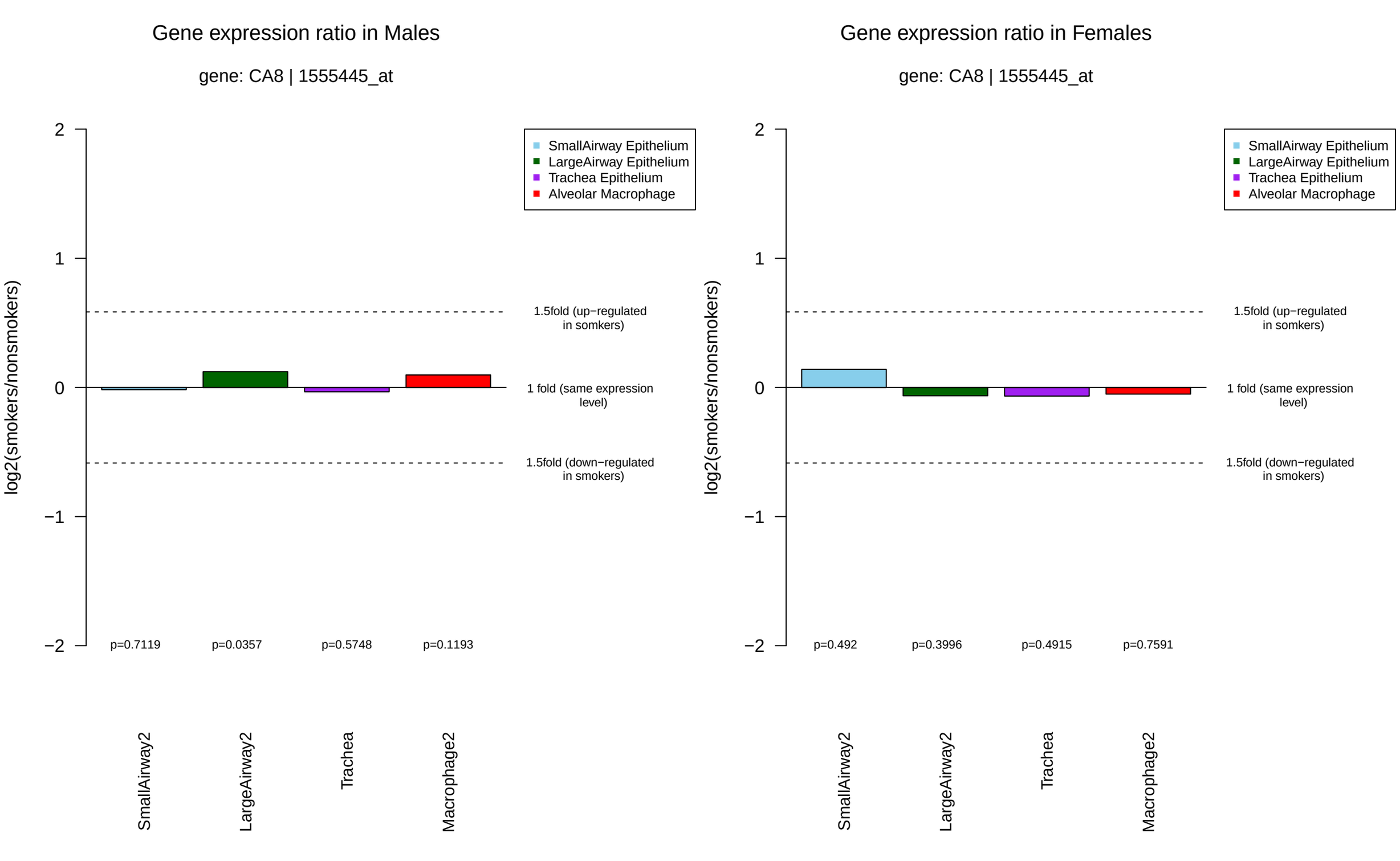
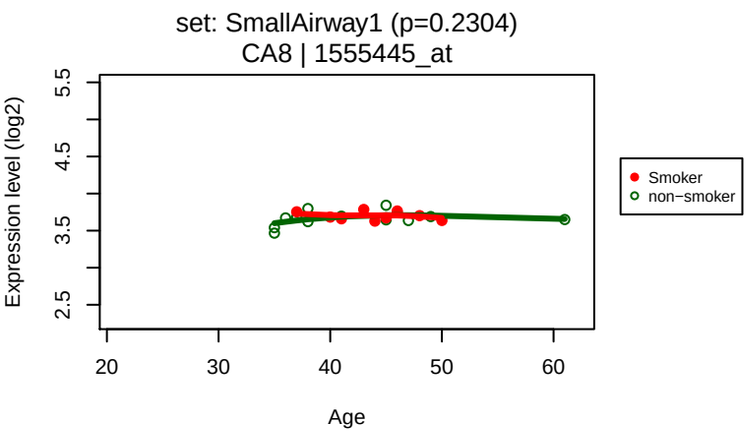
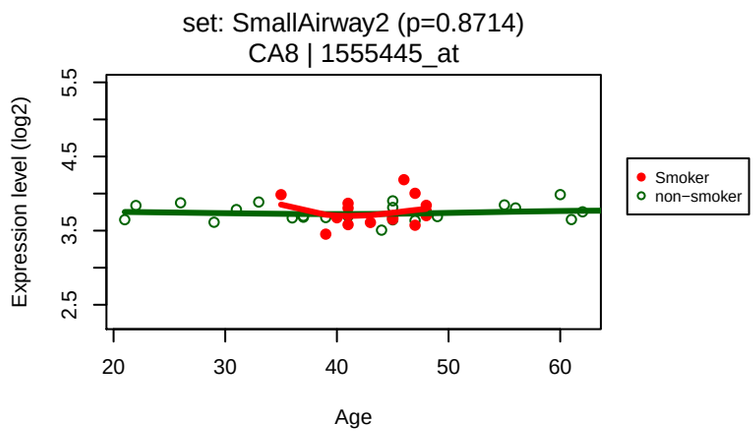
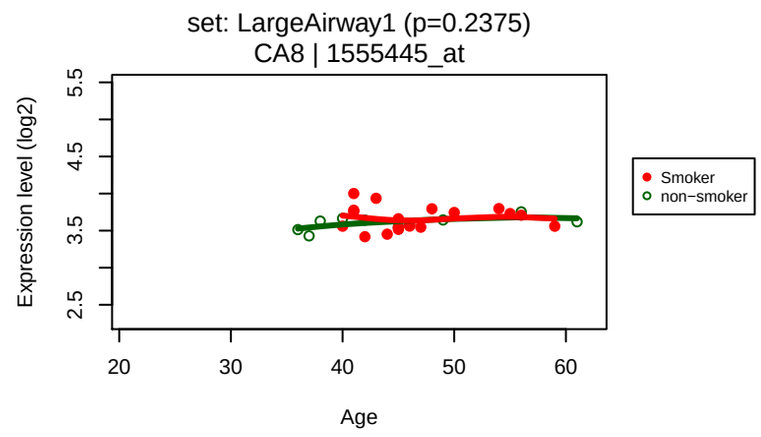

Gene expression of smokers and nonsmokers
gene: CA8 | 1555445_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2304
p=0.8714
p=0.9328
p=0.2375
p=0.2752
p=0.3177
p=0.2038
p=0.4305
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

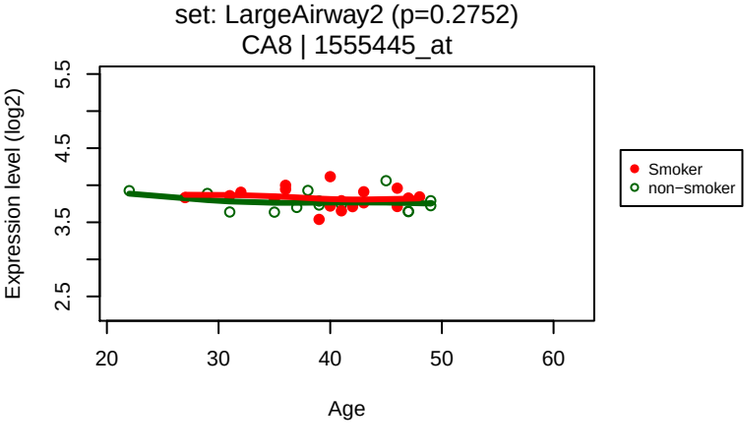

Gene expression ratio (smokers/nonsmokers)
gene: CA8 | 1555445_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2304
p=0.8714
p=0.9328
p=0.2375
p=0.2752
p=0.3177
p=0.2038
p=0.4305
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

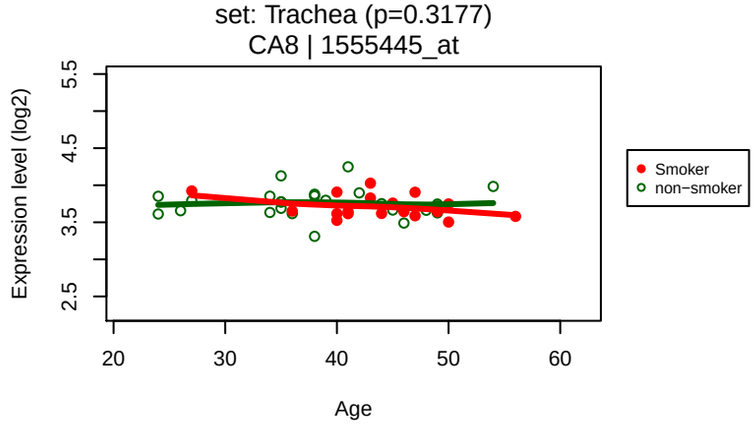

Gene expression ratio in Males
gene: CA8 | 1555445_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7119
p=0.0357
p=0.5748
p=0.1193
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CA8 | 1555445_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.492
p=0.3996
p=0.4915
p=0.7591
SmallAirway2
LargeAirway2
Trachea
Macrophage2

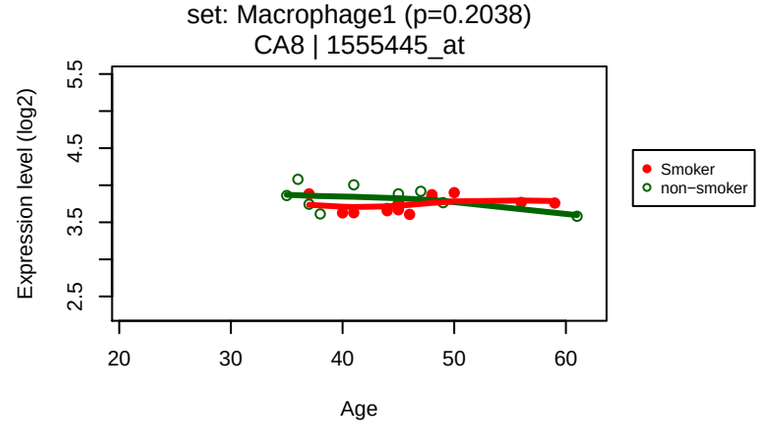

set: SmallAirway1 (p=0.2304)
CA8 | 1555445_at
Expression level (log2)
Age
Smoker
non−smoker

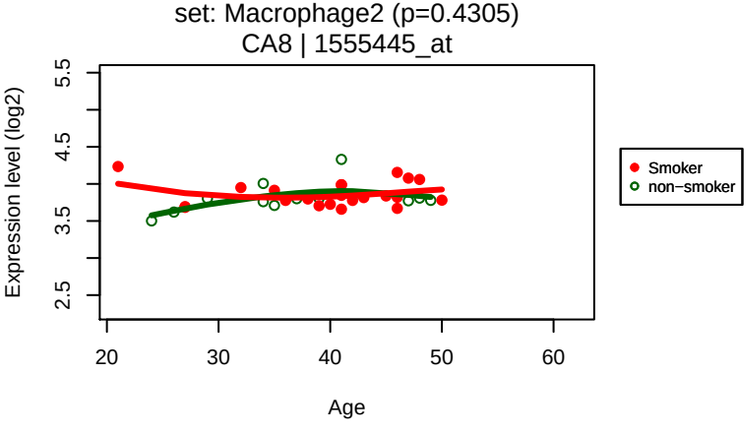

set: SmallAirway2 (p=0.8714)
CA8 | 1555445_at
Expression level (log2)
Age
Smoker
non−smoker

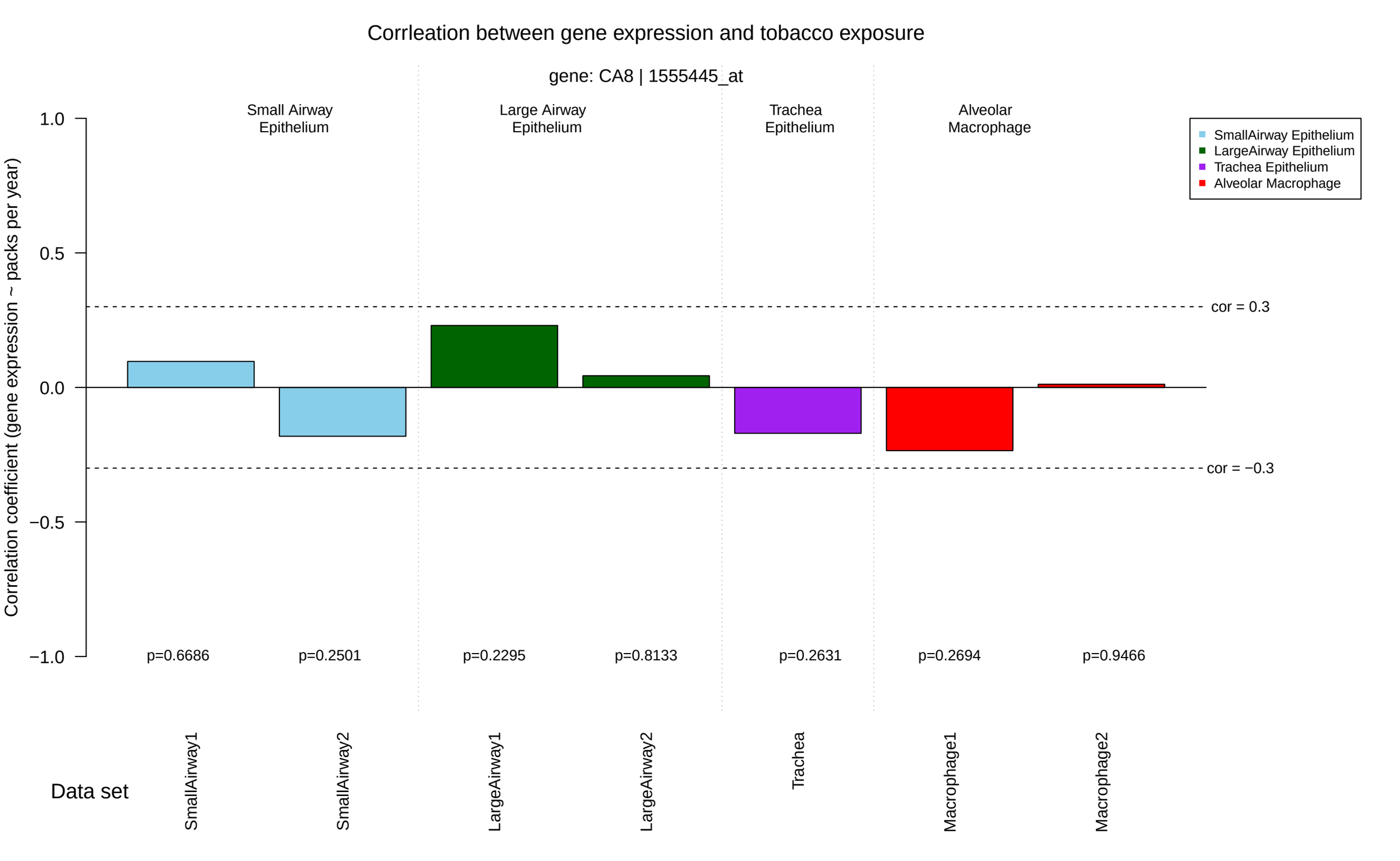

set: LargeAirway1 (p=0.2375)
CA8 | 1555445_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2752)
CA8 | 1555445_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3177)
CA8 | 1555445_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2038)
CA8 | 1555445_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4305)
CA8 | 1555445_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CA8 | 1555445_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6686
p=0.2501
p=0.2295
p=0.8133
p=0.2631
p=0.2694
p=0.9466
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2